25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Distributions & Maximum Likelihood Practice Exercises

1 MCQ

A feature in an AI vision model follows a normal distribution $X \sim \mathrm{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

A $\mu = 25,\ \sigma^2 = 36$

B $\mu = 36,\ \sigma^2 = 25$

C $\mu = 25,\ \sigma^2 = 6$

D $\mu = 6,\ \sigma^2 = 36$

2 MCQ

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\mathrm{MLE}}$?

A $\hat{\mu}_{\mathrm{MLE}} = 3$

B $\hat{\mu}_{\mathrm{MLE}} = 6$

C $\hat{\mu}_{\mathrm{MLE}} = 12$

D $\hat{\mu}_{\mathrm{MLE}} = 1.5$

3 MCQ

Given $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\mathrm{MLE}}$?

A $\hat{\mu}_{\mathrm{MLE}} = 3$

B $\hat{\mu}_{\mathrm{MLE}} = 6$

C $\hat{\mu}_{\mathrm{MLE}} = 24$

D $\hat{\mu}_{\mathrm{MLE}} = 1.5$

4 MCQ

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\mathrm{MLE}}$?

A $\hat{\mu}_{\mathrm{MLE}} = 3$

B $\hat{\mu}_{\mathrm{MLE}} = 6$

C $\hat{\mu}_{\mathrm{MLE}} = 36$

D $\hat{\mu}_{\mathrm{MLE}} = 1.5$

5 MCQ

Given $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\mathrm{MLE}}$?

A $\hat{\mu}_{\mathrm{MLE}} = 3$

B $\hat{\mu}_{\mathrm{MLE}} = 6$

C $\hat{\mu}_{\mathrm{MLE}} = 48$

D $\hat{\mu}_{\mathrm{MLE}} = 1.5$

6 MCQ

Given $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 3$

B $\hat{\mu}_{\text{MLE}} = 6$

C $\hat{\mu}_{\text{MLE}} = 60$

D $\hat{\mu}_{\text{MLE}} = 1.5$

7 MCQ

Given $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 3$

B $\hat{\mu}_{\text{MLE}} = 6$

C $\hat{\mu}_{\text{MLE}} = 72$

D $\hat{\mu}_{\text{MLE}} = 1.5$

8 MCQ

In training a language model token predictor, a binary token occurs $k = 3$ times across $n = 10$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

A $\hat{p}_{\text{MLE}} = 0.300$

B $\hat{p}_{\text{MLE}} = 0.450$

C $\hat{p}_{\text{MLE}} = 0.150$

D $\hat{p}_{\text{MLE}} = 0.700$

9 MCQ

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?

A $\frac{1}{10}$

B $\frac{3}{10}$

C $\frac{6}{10}$

D $\frac{4}{10}$

10 MCQ

Principal Component Analysis on covariance matrix Σ_A yields eigenvalues $\lambda_1 = 51$ and $\lambda_2 = 49$. Determine the percentage of variance f_x explained by the first principal component.

A 33

B 55

C 51

D 60

11 MCQ

Consider an i.i.d. sample $x_1, x_2, \ldots, x_n$ from a Poisson distribution with probability mass function $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$ for $k = 0, 1, 2, \ldots$, where $\theta > 0$. Given a sample of size $n = 5$ with sample sum $\sum_{i=1}^{5} x_i = 15$, find the maximum likelihood estimate (MLE) $\hat{\theta}$ of θ.

A 5

B 15

C 2

D 3

12 MCQ

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

A $\lambda = 1$

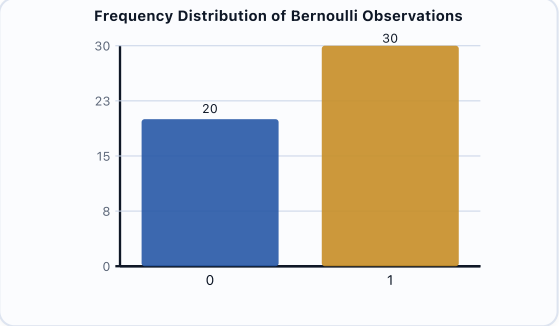
B $\lambda = e^{-3}$

C $\lambda = 3$

D $\lambda = 0$

13 MCQ

A dataset consists of observations $x_1, x_2, \ldots, x_n$ that are assumed to be drawn independently from a Bernoulli distribution with parameter p. The probability mass function is given by $P(X = k; p) = p^k(1-p)^{1-k}$ for $k \in \{0, 1\}$. The following bar chart shows the frequency distribution of these observations, where $n = 50$. Using the method of Maximum Likelihood Estimation (MLE), determine the estimated value of p for this distribution.



A $\frac{3}{5}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{2}$

14 MCQ

Consider a sequence of n independent Bernoulli trials, where the probability of success in each trial is p. Let X be the random variable representing the number of successes in these n trials. Thus, $X \sim \text{Binomial}(n, p)$. We are given a dataset of N observed values $\{x_1, x_2, \ldots, x_N\}$ for X, where each x_i is an observed count of successes from n trials. The observed data are: [3, 4, 2, 5, 4, 3, 2, 4, 5, 3]. If we assume that $n = 5$ (the number of trials for each observation is fixed at 5), find the value of p that maximizes the likelihood of observing this dataset. Then, compute the maximum likelihood estimate (MLE) of the probability of success, p, and round your answer to two decimal places.

A 0.70

B 0.75

C 0.60

D 0.67

15 MCQ

A binary classifier $f(x)$ produces the confusion matrix below with $\text{TP} = 80$, $\text{FP} = 10$, $\text{FN} = 42$, and $\text{TN} = 65$. Determine the total number of correct classifications $g = \text{TP} + \text{TN}$.

A 165

B 149

C 140

D 145

16 MCQ

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$.
Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ.
Part 2: Set the derivative to zero to find the MLE of λ.

A $\hat{\lambda} = \frac{1}{3}$

B $\hat{\lambda} = \frac{3+5+4}{3} = 4$

C $\hat{\lambda} = \frac{3+5+4}{3 \times e}$

D $\hat{\lambda} = \frac{\sum x_i}{\sum \log(x_i!)}$

17 MCQ

From the given relation $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 6$

B $\hat{\mu}_{\text{MLE}} = 3$

C $\hat{\mu}_{\text{MLE}} = 24$

D $\hat{\mu}_{\text{MLE}} = 1.5$

18 MCQ

Provided with $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 6$

B $\hat{\mu}_{\text{MLE}} = 36$

C $\hat{\mu}_{\text{MLE}} = 3$

D $\hat{\mu}_{\text{MLE}} = 1.5$

19 MCQ

Based on the condition $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 3$

B $\hat{\mu}_{\text{MLE}} = 6$

C $\hat{\mu}_{\text{MLE}} = 48$

D $\hat{\mu}_{\text{MLE}} = 1.5$

20 MCQ

Starting from $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 6$

B $\hat{\mu}_{\text{MLE}} = 60$

C $\hat{\mu}_{\text{MLE}} = 3$

D $\hat{\mu}_{\text{MLE}} = 1.5$

21 MCQ

Knowing that $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

A $\hat{\mu}_{\text{MLE}} = 3$

B $\hat{\mu}_{\text{MLE}} = 6$

C $\hat{\mu}_{\text{MLE}} = 72$

D $\hat{\mu}_{\text{MLE}} = 1.5$

22 MCQ

Considering training a language model token predictor, a binary token occurs $k = 6$ times across $n = 20$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

A $\hat{p}_{\text{MLE}} = 0.450$

B $\hat{p}_{\text{MLE}} = 0.300$

C $\hat{p}_{\text{MLE}} = 0.150$

D $\hat{p}_{\text{MLE}} = 0.700$

23 MCQ

Examining training a language model token predictor, a binary token occurs $k = 9$ times across $n = 30$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

A $\hat{p}_{\text{MLE}} = 0.450$

B $\hat{p}_{\text{MLE}} = 0.300$

C $\hat{p}_{\text{MLE}} = 0.150$

D $\hat{p}_{\text{MLE}} = 0.700$

24 MCQ

For the configuration in training a language model token predictor, a binary token occurs $k = 12$ times across $n = 40$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

A $\hat{p}_{\text{MLE}} = 0.450$

B $\hat{p}_{\text{MLE}} = 0.150$

C $\hat{p}_{\text{MLE}} = 0.300$

D $\hat{p}_{\text{MLE}} = 0.700$

25 MCQ

Analyzing training a language model token predictor, a binary token occurs $k = 15$ times across $n = 50$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

A $\hat{p}_{\text{MLE}} = 0.300$

B $\hat{p}_{\text{MLE}} = 0.450$

C $\hat{p}_{\text{MLE}} = 0.150$

D $\hat{p}_{\text{MLE}} = 0.700$